Lukas Hartmannsberger

Backend Developer, Async Python

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SUMMARY

Backend developer with eight years of experience in event-driven Python systems for mobility and IoT telemetry. I like explicit concurrency, small modules and honest estimates. Prefers measuring before optimising. I write down every design decision in short architecture notes, and I enjoy debugging unusual concurrency problems with a profiler and patience.

SKILLS

Python, asyncio, aiohttp, FastAPI, MQTT, TimescaleDB, PostgreSQL, Redis Streams, pytest-asyncio, Locust, Grafana, Protobuf, Pydantic, structlog

EXPERIENCE

Senior Backend Developer, Isar Mobility GmbH 2019 – Present

- Built an asyncio ingestion service handling 25,000 vehicle messages per second on four small instances.
- Modelled telemetry in TimescaleDB with continuous aggregates, making fleet dashboards load in under a second.
- Created load tests with Locust and documented capacity limits for the product team.
- Set up Grafana dashboards and alert rules that the whole team now uses.
- Mentor a working student and review pull requests from two other teams.

Python Developer, Bavaria Sensorik 2016 – 2019

- Developed MQTT bridges between factory devices and a Django admin portal.
- Replaced a threaded poller with asyncio, cutting memory use by 60%.
- Wrote the Protobuf schemas and versioning rules for device firmware messages.
- Debugged a rare race condition in a polling loop that had been losing roughly one message in a million.
- Presented results to the product owner every sprint with simple benchmarks.

Working Student, Software Development, Alpen Automation 2014 – 2016

- Wrote Python test tools that replayed recorded sensor data against embedded firmware.
- Created small Flask tools for the lab team to browse measurement runs.
- Documented the test procedures in the team wiki.
- Created a replay tool that fed recorded telemetry into staging, making regression testing of the ingestion service reproducible.
- Documented the message contracts and error codes for partner teams.

EDUCATION

- MSc Software Engineering, Technical University of Southern Bavaria (2014 – 2016)
- BSc Informatics, Technical University of Southern Bavaria (2011 – 2014)

LANGUAGES

German (native), English (C1)

CERTIFICATIONS

- Advanced Python Concurrency, Isar Developer Days

- Time-Series Data Modelling workshop